

Portfolio Optimization Using Machine Learning

This project explores how machine learning can be used to build a more flexible and realistic portfolio optimization framework. Traditional portfolio methods often rely on fixed rules or historical averages and usually ignore practical constraints such as taxes and individual risk preferences. The aim here is to design a system that adapts to market data while remaining grounded in real-world investment considerations.

The overall workflow follows a predict optimize evaluate pipeline. Historical price data is first processed to compute returns and simple market indicators such as volatility, trends, and momentum. These features are used by a machine learning model to estimate return signals. Instead of predicting exact future prices, the model focuses on learning relative return patterns across assets, which makes the signals more stable and suitable for portfolio construction.

Once return signals are obtained, they are passed into a constrained optimization step to compute portfolio weights. The optimization balances expected return against risk using a covariance based risk term and enforces practical constraints such as full investment and bounded position sizes. This setup is inspired by classical mean variance optimization, but uses model based return estimates instead of fixed historical averages.

A key part of the system is tax aware portfolio construction. In practice, frequent rebalancing can trigger capital gains taxes and reduce net performance. To account for this the optimization discourages unnecessary turnover and prefers existing positions unless the expected benefit justifies a trade. This allows the strategy to focus on post tax outcomes rather than raw returns.

The framework also supports personalized portfolio strategies by adjusting optimization parameters based on risk preference. This makes it possible to generate different portfolios using the same pipeline, such as:

- Conservative portfolios that prioritize stability and lower volatility
- Aggressive portfolios that accept higher risk for higher expected returns

Portfolio performance is evaluated using standard financial metrics, including:

- Annualized Return to measure overall profitability
- Volatility to quantify risk
- Sharpe Ratio for risk adjusted performance
- Maximum Drawdown to capture worst case losses

Overall this notebook represents an exploratory implementation that combines machine learning with classical portfolio optimization while incorporating tax awareness and personalization. The focus is on understanding the methodology and trade offs rather than building a production ready trading system.